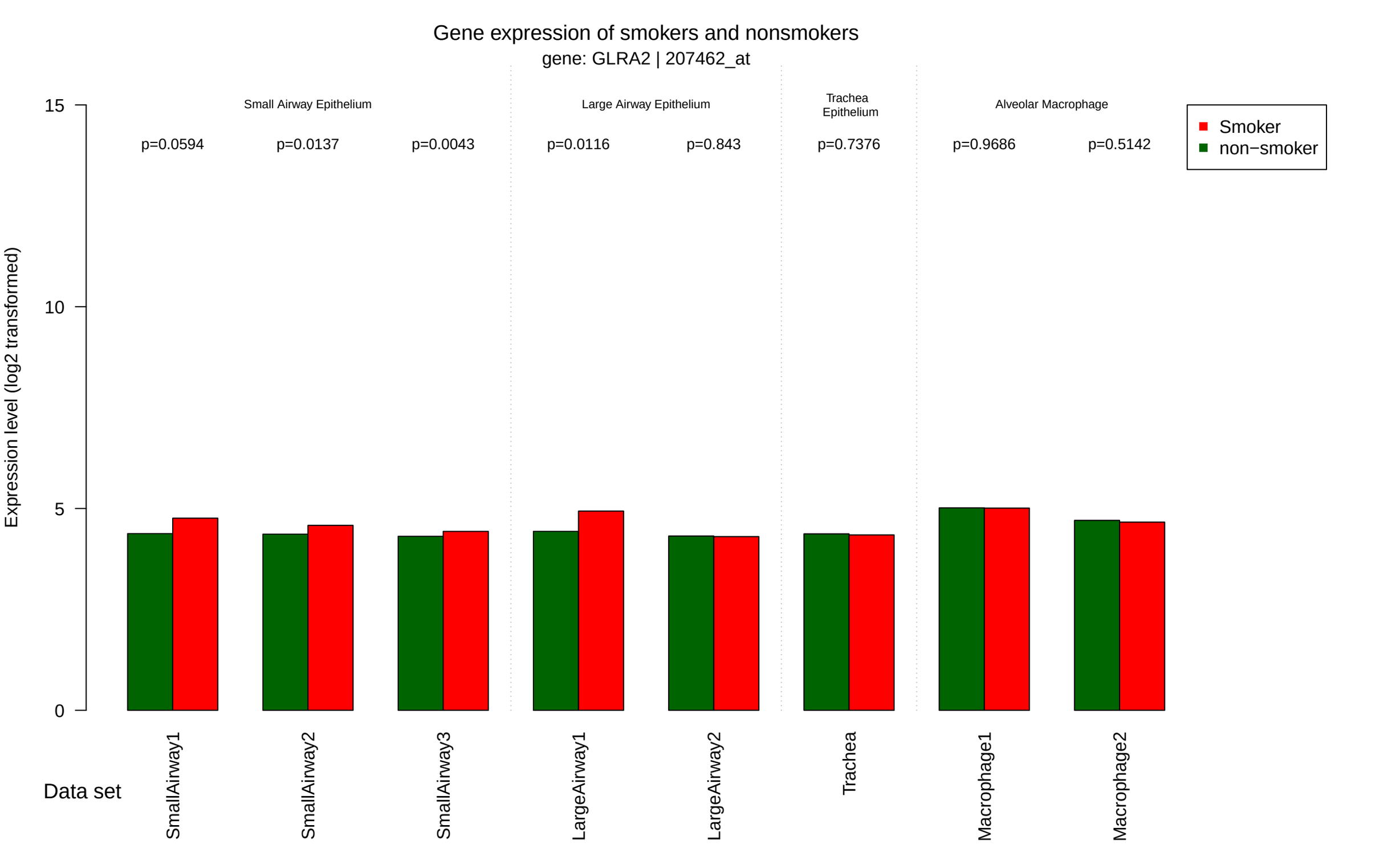
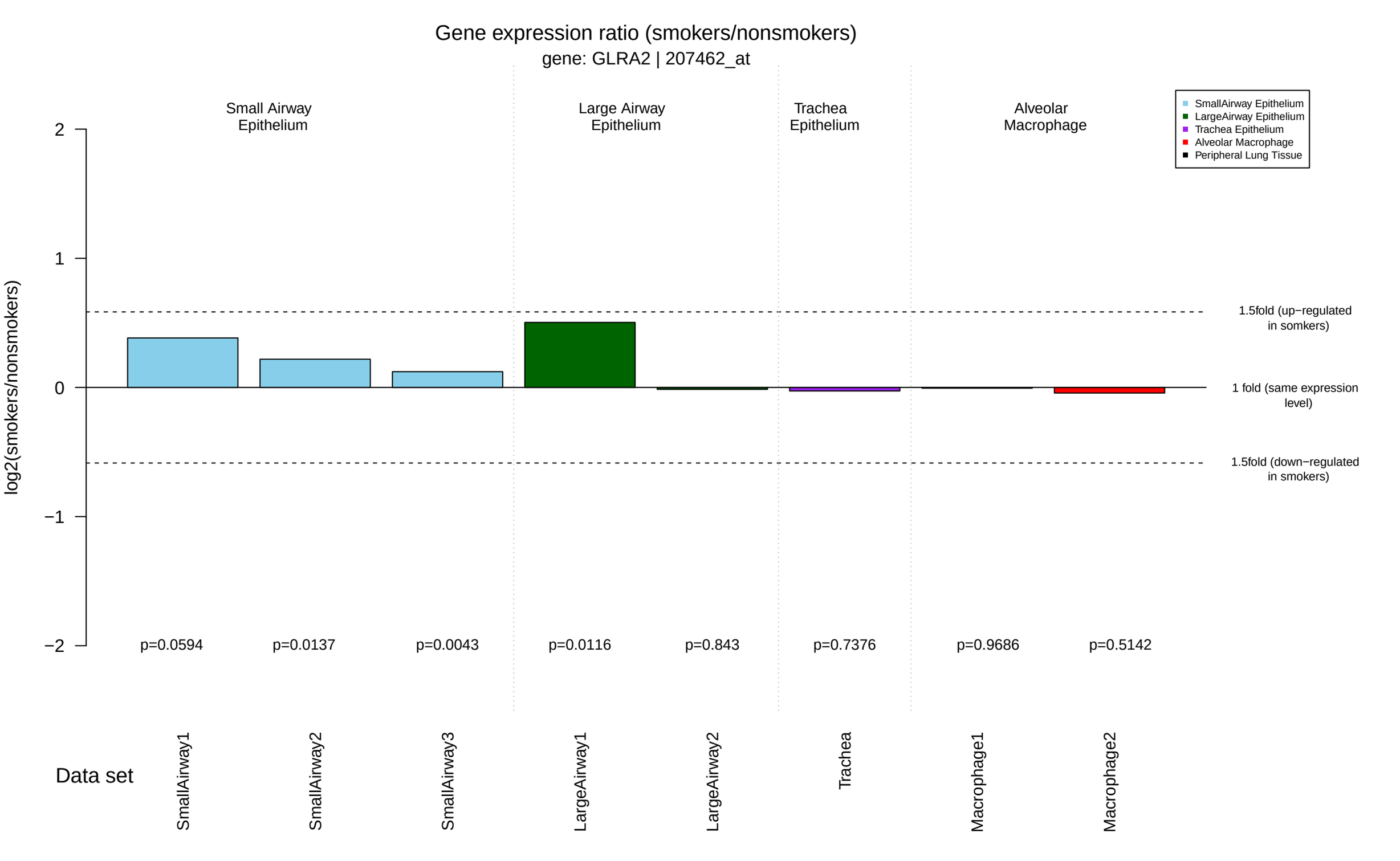
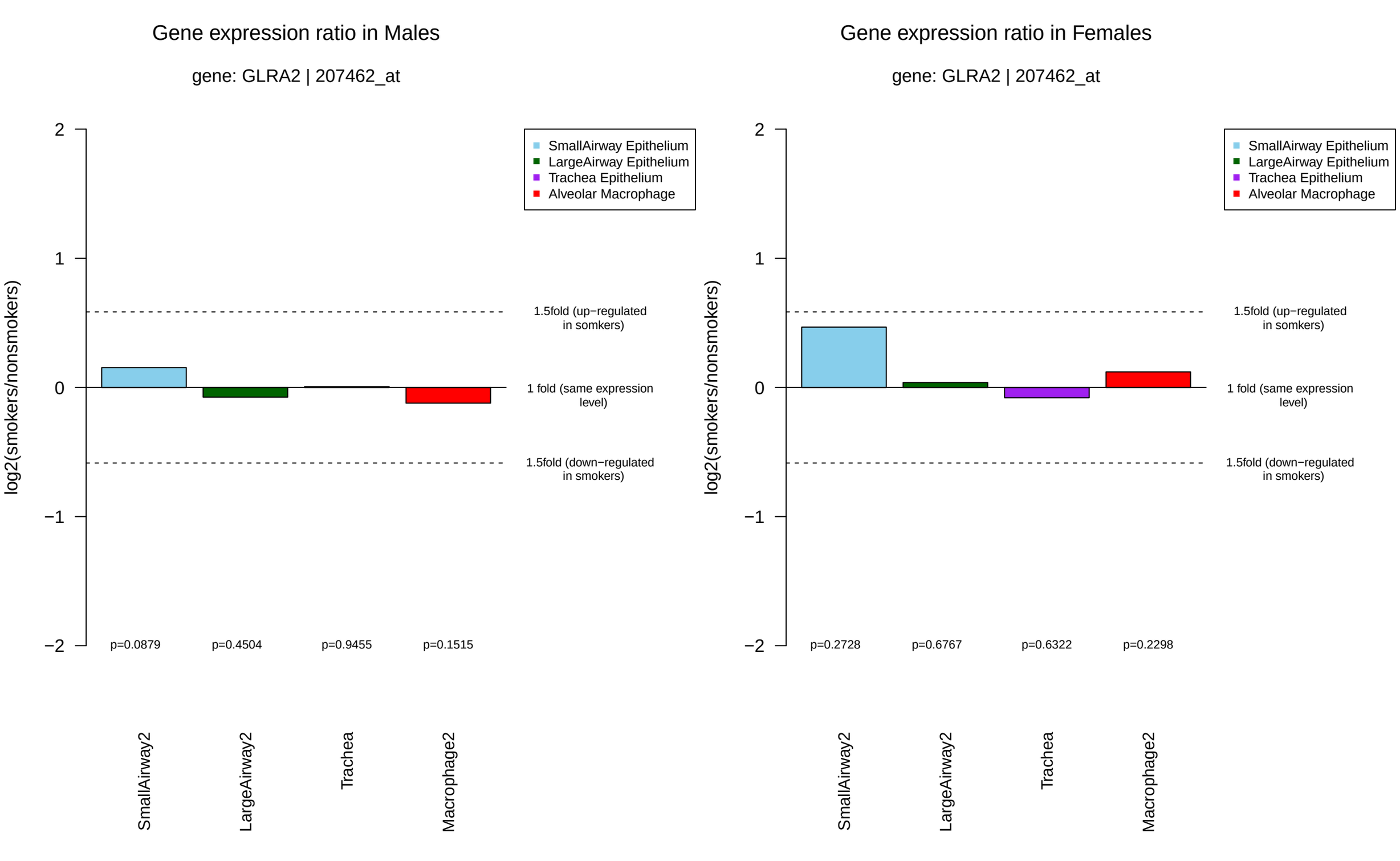
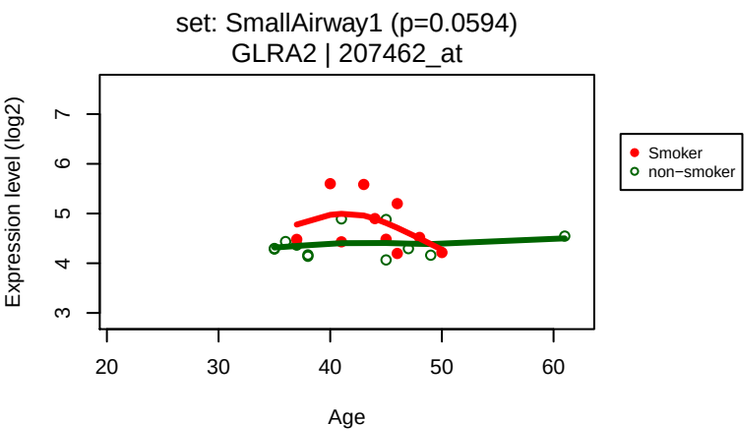
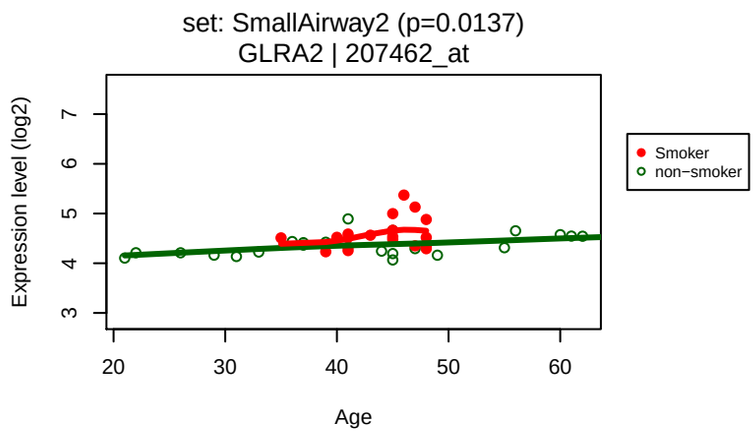
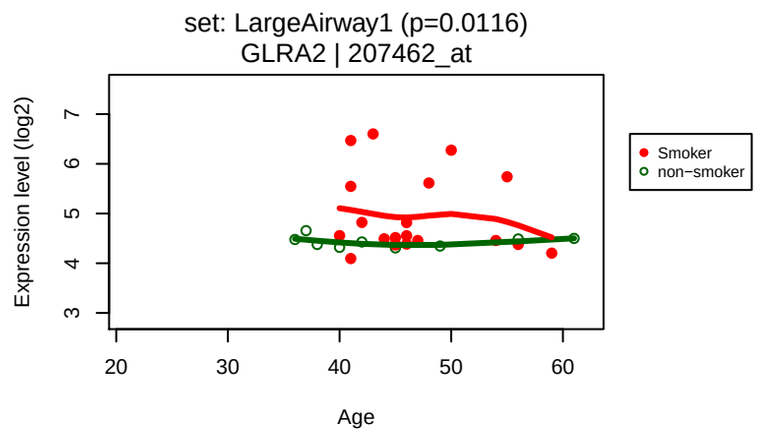

Gene expression of smokers and nonsmokers
gene: GLRA2 | 207462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0594
p=0.0137
p=0.0043
p=0.0116
p=0.843
p=0.7376
p=0.9686
p=0.5142
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

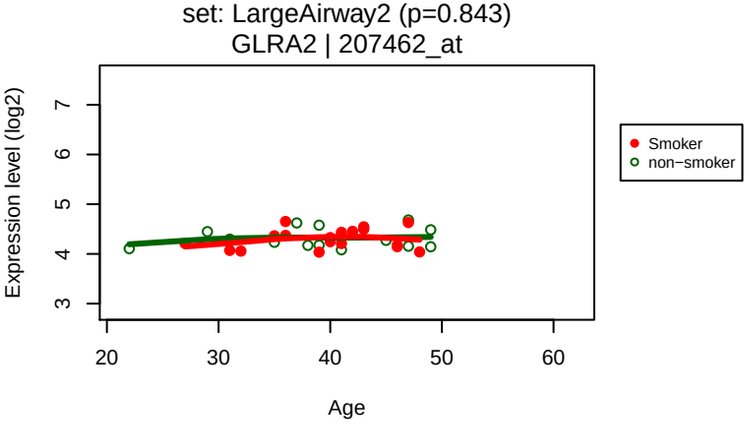

Gene expression ratio (smokers/nonsmokers)
gene: GLRA2 | 207462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0594
p=0.0137
p=0.0043
p=0.0116
p=0.843
p=0.7376
p=0.9686
p=0.5142
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

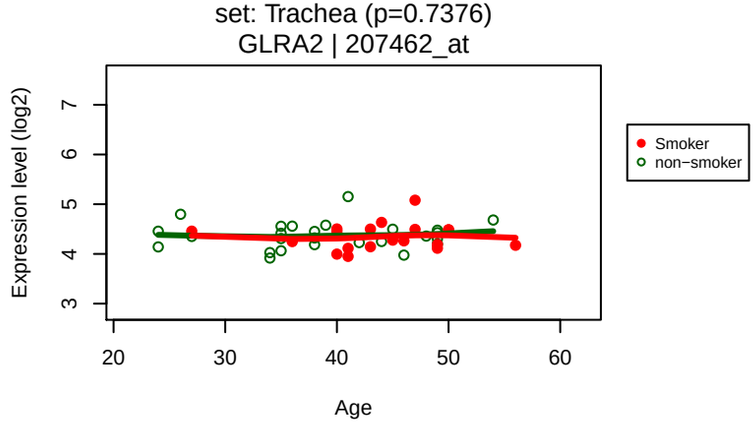

Gene expression ratio in Males
gene: GLRA2 | 207462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0879
p=0.4504
p=0.9455
p=0.1515
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GLRA2 | 207462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2728
p=0.6767
p=0.6322
p=0.2298
SmallAirway2
LargeAirway2
Trachea
Macrophage2

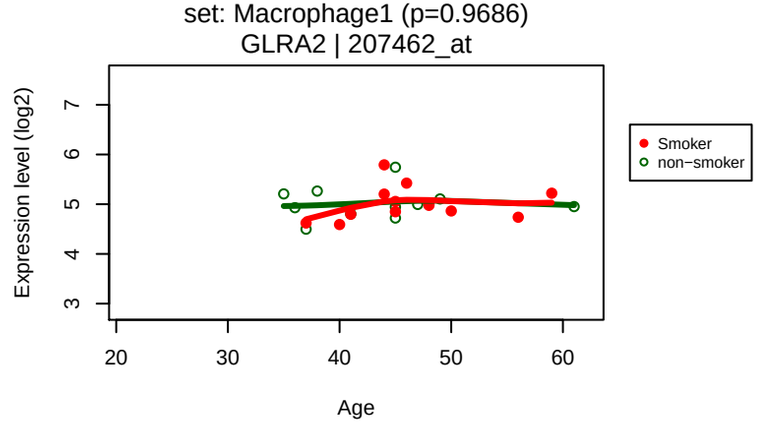

set: SmallAirway1 (p=0.0594)
GLRA2 | 207462_at
Expression level (log2)
Age
Smoker
non−smoker

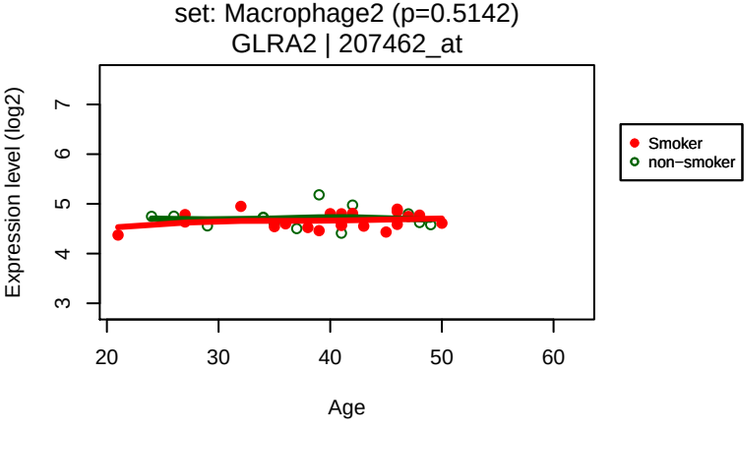

set: SmallAirway2 (p=0.0137)
GLRA2 | 207462_at
Expression level (log2)
Age
Smoker
non−smoker

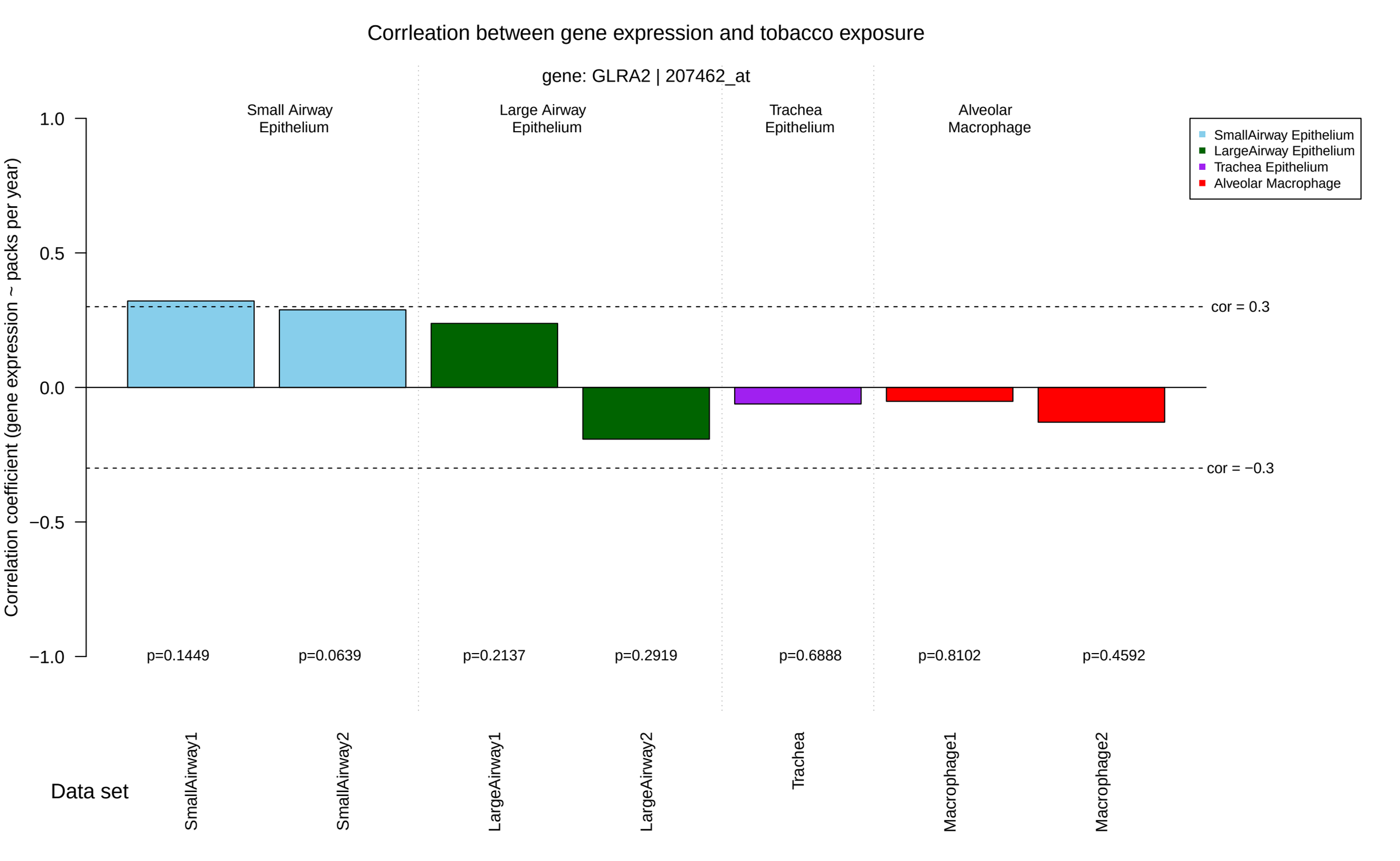

set: LargeAirway1 (p=0.0116)
GLRA2 | 207462_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.843)
GLRA2 | 207462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7376)
GLRA2 | 207462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9686)
GLRA2 | 207462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5142)
GLRA2 | 207462_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GLRA2 | 207462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1449
p=0.0639
p=0.2137
p=0.2919
p=0.6888
p=0.8102
p=0.4592
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2